

ASSIGNMENT - 2

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Q. Explain the Addressing format of IPv4 & IPv6.
Discuss any 3 major differences b/w them.

Ans. IPv4 Addressing Format :-

- Size - 32 bits
- Representation - 4 decimal no. separated by dots
- Range of each block - 0-255
- Example - 192.168.0.1
- Structure - IPv4 addresses are divided into:
 - : Network portion: identifies network
 - : Host portion: identifies device within that network.

Subnetting using masks defines how many bits belong to each portion

IPv6 Addressing Format :-

- Size - 128 bits
- Representation - 8 groups of 4 hexadecimal digits separated by colons
- Example - 2001:0db8:85a3:0000:78ab:8a26:0370:7344
- Structure - IPv6 addresses include diff. types such as:
 - : Global unicast
 - : Link-local

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- : Multi cast
- : No broadcast addresses.

Major Differences

IPv4	IPv6
• 32 bit address space • Represented in Decimal format, separated by Dots. • Often requires Network address Translation (NAT) due to limited addresses, supports DHCP for configuration	• 128-bit address space • Hexadecimal format, separated by colons • No need for NAT, supports stateless address configuration, make device self-configure automatically

Q3. Describe the purpose of ARP, RARP, BOOTP & DHCP. How do these protocols help in assigning or resolving network addresses?

Ans. 1. ARP :-

purpose → It's used to map an IPv4 address to a physical address on local network

network address →

- ARP doesn't assign IP addresses.
- It resolves an existing IPv4 addresses to its corresponding MAC address.
- This ^{is} required for local communication on LAN.

2. RARP :-

Purpose → It performs the opposite funⁿ of ARP. It allows device to discover its IPv4 address from a known MAC address.

network address →

- It helps a device learn its IP address when it only knows its MAC address.

- Used by older diskless computers that had no IP stored.

3. BOOTP :-

Purpose → It's used to automatically assign an IP address to device, especially during booting & provide additional configuration parameters.

network address →

- It provides static IP address to device at boot time.

- The server contains a table that maps each device's MAC address to specific IP address.
- Also sends network parameters.

4. DHCP :-

Purpose → It's more advanced version of BOOTP. It dynamically assigns IP addresses & other network configuration details to devices.

- network addresses →
- It automatically assigns IP addresses to devices on network.
 - Can provide dynamic or reserved addresses.
 - also supplies subnet mask, gateway, DNS, lease time, etc.
 - Uses the DORA process.

Q. Explain direct vs indirect delivery. Also describe how a router performs packet forwarding using its forwarding table.

Ans. # Direct Delivery → It happens when the source & destination are on same physical networks.

How it works :-

- Host checks destination IP address.
- It uses its subnet mask to determine the network portion.

- If destination is on same network, host sends packet directly to destination.
- Host uses ARP to find destination's MAC address.

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Example - Source IP: 192.168.1.10
Destination IP: 192.168.1.20

Indirect Delivery → It occurs when destination is on diff. network, so packet must be forwarded via router.

Now it works :-

- Host determines destination is outside its subnet.

- Sends the packet to router's MAC address, not the final destination's MAC.
- The router then takes packet & forwards it towards its destination network.

Example - Source IP: 192.168.1.10
Destination IP: 10.0.0.8

Packet forwarding Using Forwarding Table →

Step 1 :- Receive Packet : router receives an IP packet on one of its interfaces.

Step 2 :- Extract destination IP : router reads packet's destination IP address

Step 3 :- Perform "Longest Prefix Match" : router looks through its forwarding table to find most specific route.

Step 4:- Determine Next Hop: Routing table entry tells the router what to do -

- next hop address (another router), or
- Directly connected network

Step 5:- Use ARP (if needed): Uses ARP to find MAC address of next hop.

Step 6:- Forward the packet: The router encapsulates the packet in new frame with next hop MAC address & sends it through appropriate interface.

Q4. Describe working ~~principle~~ principles of any 2 unicast routing protocols (RIP, OSPF or BGP). Mention their routing metrics & basic characteristics.

Ans. # Open Shortest Path First (OSPF):- It's a modern Link State routing approach.

Working Principle → Each router builds a complete map of the network using Link State Advertisements (LSAs).

- Router flood LSAs within an area.
- Every router independently computes the shortest path tree using Dijkstra's algo.

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- Routing is hierarchical using areas (Area 0 is backbone).

Routing Metric → • Cost : - Derived from interface bandwidth
- Higher bandwidth = lower cost
- Formula $\Rightarrow \text{cost} = \frac{10^8}{\text{Bandwidth (bps)}}$

Basic Characteristics → • Type = Link State protocol
• Autonomous System = IGP
• Algorithm = Dijkstra (SPF)
• Convergence Speed = Fast
• Scalability = High
• Features = * Multicast for updates
* Supports area, authentication, VLSM

Routing Information Protocol (RIP) :- It's one of the oldest

Interior Gateway Protocols (IGP). It uses Distance vector routing approach.

Working Principle → • Each router periodically advertises its routing table to its neighbors
• Routers exchange info about distance to other networks.
• Each router selects the path with least hop count.
• Uses Bellman-Ford algo for path computation
• updates are broadcast or multicast.

Routing metrics → • Hop Count: - Each router hop increases the metric by 1.
• Max. hop count = 15 (16 = Unreachable).

Basic Characteristics → • Autonomous System: IGP
• Algo: Bellman-Ford
• Type: Distance Vector Protocol
• Convergence Speed: slow
• Scalability: Poor
• Versions: ~~RIP v1, RIP v2, RIPv3~~

Q5 Compare UDP, TCP & SCTP in terms of reliability, Connection Type & typical applications. Provide examples where each protocol is used.

Ans. # UDP :-
• Unreliable
• Connection Less.
• Best ^{for} applications where:
- Speed is more imp. than reliability
- Data is tolerant of loss or already handles reliability at application level

Examples → • DNS • VoIP • DHCP
• Online gaming • Real-time applications

TCP:-
• reliable
• Connection-oriented (3-way)
• Used where accuracy & completeness are

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critical

Examples →

- HTTP/HTTPS
- Email
- File Transfer
- Remote Login

SCTP (Stream Control Transmission Protocol) :-

- Reliable
- Connection-oriented (4-way)
- Used where multiple streams must be carried without block.
- Used where high reliability + fault tolerance are needed.

Example →

- Telecommunication Signaling
- Mission Critical application
- Industrial Control system
- WebRTC

Q6. Explain how transport layer supports process-to-process communication. What is the role of port no. in this process?

Ans. The transport layer is responsible for delivering data from one specific process on source host to a specific process on destination host.

Networking below transport layer only provides host-to-host delivery.

The transport layer enables process-to-process communication by:

1. Identifying which application/process should receive data.
2. Ensuring correct delivery using mechanisms like:
 - Segmentation & reassembly
 - error control
 - connection establishment
 - Multiplexing & Demultiplexing.

Role of Port Numbers → Port no. are identifiers used to distinguish diff. processes running on the same host.

- Each application using network is assigned a port no.
- Transport layer uses 2 key identifiers:-
 1. Source Port No. :- identifies the sending process on source device.
 - helps in sending replies back to correct application.
 2. Destination Port No. :- identifies which application on destination computer should receive data.

Q7. What is Congestion Control? Explain any 2 techniques used in TCP Congestion Control (e.g. slow start, Congestion Avoidance, Fast Retransmit)

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Ans: Congestion Control $\rightarrow$ It refers to techniques used in networks to prevent too much data from being injected into network, which can overload routers, increase delays & cause packet loss.

In TCP, congestion control ensures that the sending rate is adjusted according to the network capacity. TCP uses variable called congestion window, which determines how much data can be sent before waiting for acknowledgements.

TCP Congestion Control Techniques $\rightarrow$

1. slow start :- It's used when
 - TCP connection begins
 - After timeout

Working Principle :- TCP starts with a very small congestion window, usually 1 MSS

- For every ACK received, cwnd increases exponentially
 $\rightarrow cwnd = cwnd * 2$ each RTT

It allows TCP to probe network capacity carefully but quickly.

When slow start ends $\rightarrow$

- when cwnd reaches a threshold called ssthresh

- when packet loss occurs.

2. Congestion Avoidance :- It takes over after slow start phase to avoid overloading network.

Working Principle → • Growth of cwnd become linear, not exponential.

• For every RTT, cwnd increases by 1 MSS.

It helps maintain network stability, Linear increase prevents congestion in networks that are already near capacity.

When Packet loss is Detected → • ssthresh is set to half of current cwnd.

• cwnd is reduced.

Q8. Explain Concept of Quality of Service (QoS). Describe working of Leaky Bucket & Token Bucket algo & state one difference b/w them.

Ans. QoS refers to set of techniques used in networking to manage Bandwidth, reduce delay, prevent packet loss & ensure consistent performance for different applications.

QoS ensures :-

- Bandwidth allocation for imp. traffic
- Low delay for real-time services
- Low jitter for audio/video streams
- Low packet loss

Leaky Bucket Algo → It shapes traffic to make it smooth & constant.

Working Principle:

- Packets enter "bucket" at any speed,
- They are sent out at constant, fixed rate.

- If incoming packets exceed the bucket capacity, extra packets are discarded.
- The O/P traffic becomes steady & regulated.

Key Idea: Controls O/P rate strictly → prevents bursts.

Token Bucket Algo → It allows traffic to be sent in bursts as long as enough tokens are available.

Working principle:

- Tokens are added to bucket at a steady rate

- To send a packet, Sender must remove tokens equal to packet size.
- If enough tokens exist → burst is allowed.
- If not enough tokens → packet may wait or be dropped.
- Unlike the leaky bucket, if tokens accumulate, the sender can transmit, multiple packets at once.

Key Idea: Regulates avg rate but still permits bursts when tokens are available.

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Leaky Bucket

Produces a fixed, smooth O/P rate & does not allow bursts

Token Bucket

Allows bursty O/P enough token exists.

Q9 Write short notes on any 3 of the following:

- DNS • FTP • HTTP • Email protocols • SNMP

Ans. 1. DNS (Domain name System) →

It's a distributed, hierarchical naming system that translates human ~~not~~ readable domain names into IP addresses.

It works using network of DNS servers such as:

- Root Servers • Top-level Domain Servers
- Authoritative servers

DNS uses UDP port 53 for most queries & TCP port 53 for zone transfers. It improves usability by allowing people to use names instead of numeric IP addresses.

2. FTP (File Transfer Protocol) →

It's a standard network protocol used to transfer files b/w client & server. It supports:

- uploading & downloading files • Directory listing
- Authentication using username / password

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FTP uses:

- TCP port 21 for control commands.
- TCP port 20 for data transfer.

Although widely used, FTP isn't secure bcoz data including passwords are sent in plaintext.

3. HTTP (Hypertext Transfer Protocol) →

It's an application layer protocol used for transferring web documents such as HTML pages, images & scripts.

It follows request-response model where:

- Client sends an HTTP request.
- web server returns an HTTP response.

Characteristics:

- Stateless (each request is independent)
- Runs on TCP port 80.

Q10. What is firewall? Explain its purpose & any 2 types of firewalls. Also write basic idea of cryptography with 1 example each of symmetric & asymmetric encryption.

Ans. # A firewall is network security device or software that monitors & controls incoming & outgoing network traffic based on predetermined security rules. It acts as barrier b/w trusted internal network & an untrusted external network.

- Purpose :**
- Protects networks from unauthorized access.
 - Blocks malicious traffic such as viruses, worms & hackers.
 - Filters traffic based on IP addresses, ports, protocols.
 - Enforces security policies.

Type of Firewall :

1. **Packet filtering Firewall** →
 - Pros : Fast, Simple
 - Cons : Can't inspect payload, limited protection.
 - Inspects packets individually
 - Allows or blocks packets based on simple rules.
2. **Stateful Inspection Firewall** →
 - Monitors state of active connections.
 - Allows packets only if they are part of an established connection.
 - Pros : more secure than simple packet filtering
 - Cons : slightly slower due to connection tracking.

Cryptography is practice of secure communication in the presence of adversaries, using mathematical techniques to encrypt data & decrypt it only for authorized users.

- Purpose :**
- Ensures confidentiality of data
 - Ensure integrity
 - Provides authentication

- Supports non-repudiation

Types of Cryptography:

1. Symmetric Encryption →
 - Same key used for encryption & decryption
 - Example algo ⇒ Advanced Encryption Standard
 - Example use ⇒ Encrypting files on disk or secure VPN tunnels.
2. Asymmetric Encryption →
 - Use public key to encrypt & private key to decrypt
 - Example algo ⇒ RSA
 - Example use ⇒ secure email (PGP), SSL/TLS for HTTPS.

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